Relational Retrieval: Leveraging Known-Novel Interactions for Generalized Category Discovery

Yulin Xu*
University of California, Irvine
Irvine, CA, USA
yulinx8@uci.edu

Chunqi Guo*
Sichuan Agricultural University
Chengdu, China
guochunqi02@gmail.com

Yuanzhen Shuai
University College London
London, UK
ucapys0@ucl.ac.uk

Jianyuan Ni†
Juniata College
Huntingdon, PA, USA
jni100@juniata.edu

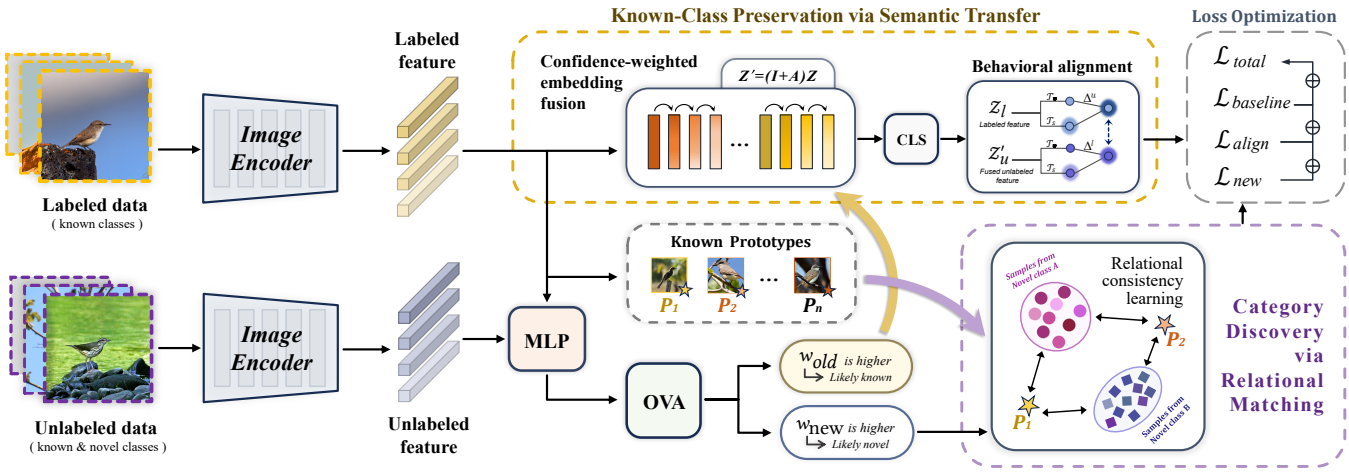

Figure 1: Overview of RPC. Soft ID/OOD decomposition via OVA classifiers enables bidirectional knowledge transfer: embedding fusion for known-class preservation and relational matching for category discovery.

Abstract

Generalized Category Discovery (GCD) aims to classify unlabeled samples into both known and novel categories, given labeled data of known classes. Existing methods treat labeled and unlabeled data as separate learning streams—supervised for labeled, self-supervised for unlabeled—overlooking the potential for mutual enhancement. To address this, we tackle GCD from a novel *relational retrieval* perspective and propose Relational Pattern Consistency (RPC), a framework that explicitly couples labeled and unlabeled data through bidirectional knowledge transfer. RPC employs One-vs-All classifiers for soft ID/OOD decomposition, then introduces two mechanisms: (i) confidence-weighted embedding fusion with behavioral alignment to transfer semantic patterns from labeled to unlabeled known-class samples, and (ii) relational consistency learning that discovers novel categories through their invariant similarity patterns with known-class prototypes, transforming unreliable pseudo-labeling into well-defined relational pattern matching. Experiments on six benchmarks demonstrate state-of-the-art performance, with RPC improving Old ACC by 1.4% on CUB over the strongest baseline while achieving strong gains on fine-grained tasks. Code will be released upon acceptance.

CCS Concepts

• **Computing methodologies → Visual content-based indexing and retrieval**.

Keywords

Generalized category discovery, semi-supervised learning, contrastive learning.

ACM Reference Format:
Yulin Xu, Chunqi Guo, Yuanzhen Shuai, and Jianyuan Ni. 2026. Relational Retrieval: Leveraging Known-Novel Interactions for Generalized Category Discovery. In *Proceedings of the 2026 ACM International Conference on Multimedia Retrieval (ICMR '26), June 16-19, 2026, Amsterdam, Netherlands.* ACM, New York, NY, USA, 5 pages. https://doi.org/XXXXXXX.XXXXXXX

*Both authors contributed equally to this research.
†Correspondence to Jianyuan Ni {jni100@juniata.edu}.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

ICMR '26, Amsterdam, Netherlands

© 2026 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-XXXX-X/2026/06
https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Generalized Category Discovery (GCD) [8, 28] extends traditional classification to open-world scenarios by jointly recognizing known and discovering novel classes in unlabeled data, with applications spanning natural images [13, 18, 24, 29, 33, 37, 38], medical imaging [4, 21, 39], industrial scenarios [9], face recognition [16, 22], and information retrieval [25].

Recent GCD methods [3, 5, 14, 15, 23, 32, 35, 36, 40] have advanced through parametric classifiers [32], spatial prompt tuning [30], textual integration [29, 38], and soft ID/OOD decomposition [14]. However, these approaches fundamentally treat labeled and unlabeled data as separate learning streams—labeled data undergoes supervised learning while unlabeled data relies on self-supervised signals. This separation overlooks a critical opportunity: labeled and unlabeled data can mutually enhance each other through structured interactions. We identify two key limitations. First, unlabeled known-class samples rely solely on self-supervision [2], despite labeled samples of the same classes having already captured highly discriminative features—wasting valuable semantic knowledge. Second, category discovery ignores relational patterns—samples from the same novel category maintain consistent relationships with known-class prototypes that serve as powerful grouping signals.

We propose **Relational Pattern Consistency (RPC)**, a framework that explicitly couples labeled and unlabeled data through bidirectional knowledge transfer (Fig. 1). RPC leverages soft ID/OOD decomposition from One-vs-All classifiers to enable targeted interactions: (i) for known-class preservation, we transfer semantic patterns from labeled to unlabeled data via confidence-weighted embedding fusion and enforce behavioral alignment under augmentation [26]; (ii) for category discovery, we introduce relational consistency—the principle that same-category samples maintain invariant relationships with known-class prototypes. This transforms discovery from ill-posed clustering into well-defined pattern matching within the known-class reference frame, which we term *relational retrieval*.

Our contributions are: (i) We propose RPC, the first GCD framework coupling labeled and unlabeled data through bidirectional knowledge transfer via soft ID/OOD decomposition. (ii) We introduce confidence-weighted embedding fusion with behavioral alignment for known-class preservation, and relational consistency learning that discovers novel categories through their invariant relationships with known-class prototypes, going beyond unreliable pseudo-labels. (iii) Experiments demonstrate state-of-the-art performance on six benchmarks, with consistent improvements in both known and novel class accuracy.

2 Problem Setting

Let $\mathcal{D}$ denote the dataset, split into a labeled part $\mathcal{D}_L = \{(\mathbf{x}_i^l, y_i^l)\} \subset \mathcal{X} \times \mathcal{Y}_l$ and an unlabeled part $\mathcal{D}_U = \{\mathbf{x}_i^u\} \subset \mathcal{X}$, where each $\mathbf{x}_i^u$ is associated with a hidden ground-truth label $\hat{y}_i \in \mathcal{Y}_u$ used only for evaluation. The unlabeled pool may contain categories absent from the labeled subset, i.e., $\mathcal{Y}_l \subset \mathcal{Y}_u$. The learning objective is to assign every instance in $\mathcal{D}_U$ to a semantic category. Following [32], we assume the total number of categories in the unlabeled set is known a priori, and write $K = |\mathcal{Y}_u|$.

3 Method

3.1 Baseline for GCD

Our model builds on SimGCD [32] for representation learning and parametric classification, while incorporating DebGCD's [14] One-vs-All (OVA) decomposition to identify whether unlabeled samples belong to known or novel classes. This combination provides reliable soft weights for our bidirectional knowledge transfer.

3.1.1 Representation Learning. The goal is to learn discriminative features for effective classification. A pre-trained feature extractor $f(\cdot)$ is fine-tuned with two contrastive losses. Given two augmentations $\hat{\mathbf{x}}_i$ and $\tilde{\mathbf{x}}_i$ of an image $\mathbf{x}_i$ in a mini-batch $\mathcal{B}$ containing both labeled and unlabeled data, the self-supervised contrastive loss is:

$$\mathcal{L}_{\text{rep}}^u = \frac{1}{|\mathcal{B}|} \sum_{\mathbf{x}_i \in \mathcal{B}} -\log \frac{\exp(\hat{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_i / \tau_u)}{\sum_{\mathbf{x}_j \in \mathcal{B}} \exp(\hat{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_j / \tau_u)}, \tag{1}$$

where $\mathbf{z}_i = g(f(\mathbf{x}_i))$ is the projected feature for contrastive learning, $g(\cdot)$ is the projection head, and τ_u is a temperature value. A supervised contrastive loss is similarly defined by:

$$\mathcal{L}_{\text{rep}}^s = \frac{1}{|\mathcal{B}_l|} \sum_{\mathbf{x}_i \in \mathcal{B}_l} \frac{1}{|\mathcal{N}_i|} \sum_{p \in \mathcal{N}_i} -\log \frac{\exp(\hat{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_p / \tau_s)}{\sum_{n \neq i} \exp(\hat{\mathbf{z}}_i^\top \tilde{\mathbf{z}}_n / \tau_s)}, \tag{2}$$

where $\mathcal{N}_i$ is the indexes of images in the mini-batch that have the same label to $\mathbf{x}_i$, $\mathcal{B}_l$ is the mini-batch of labeled training data and τ_s is a temperature value. The two losses are combined to learn the representation: $\mathcal{L}_{\text{rep}} = (1-\lambda)\mathcal{L}_{\text{rep}}^u + \lambda \mathcal{L}_{\text{rep}}^s$, where λ is a hyperparameter.

3.1.2 Classifier Learning. Following SimGCD [32], we employ a parametric classifier with learnable prototypes $\mathcal{T} = \{\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_K\}$, where $K = |\mathcal{Y}_u|$ is assumed known a priori. During training, the soft label $\hat{p}_i^k$ for each augmented view $\hat{\mathbf{x}}_i$ is computed via softmax over cosine similarities between features and prototypes:

$$\hat{p}_i^k = \frac{\exp\left((\hat{\mathbf{h}}_i / \|\hat{\mathbf{h}}_i\|_2)^\top (\mathbf{t}_k / \|\mathbf{t}_k\|_2) / \tau_s\right)}{\sum_j \exp\left((\hat{\mathbf{h}}_i / \|\hat{\mathbf{h}}_i\|_2)^\top (\mathbf{t}_j / \|\mathbf{t}_j\|_2) / \tau_s\right)}, \tag{3}$$

where $\hat{\mathbf{h}}_i = f(\hat{\mathbf{x}}_i)$. Similarly computing $\tilde{\mathbf{p}}_i$ for $\tilde{\mathbf{x}}_i$, the classifier losses become:

$$\mathcal{L}_{\text{cls}}^s = \frac{1}{|\mathcal{B}_l|} \sum_{\mathbf{x}_i \in \mathcal{B}_l} \ell_{CE}(\mathbf{y}(\mathbf{x}_i), \hat{\mathbf{p}}_i), \quad \mathcal{L}_{\text{cls}}^u = \frac{1}{|\mathcal{B}|} \sum_{\mathbf{x}_i \in \mathcal{B}} \ell_{CE}(\tilde{\mathbf{p}}_i, \hat{\mathbf{p}}_i), \tag{4}$$

where $\mathbf{y}_i$ is the ground truth label for the labeled data point $\mathbf{x}_i$, ℓ_{CE} is the cross-entropy loss, and $H(\mathbf{p}) = -\sum \mathbf{p} \log \mathbf{p}$ regularizes the mean prediction $\mathbf{p} = \frac{1}{2|\mathcal{B}|} \sum_{\mathbf{x}_i \in \mathcal{B}} (\hat{\mathbf{p}}_i + \tilde{\mathbf{p}}_i)$ in a mini-batch. The final classifier loss is given by $\mathcal{L}_{\text{cls}} = (1-\lambda)\mathcal{L}_{\text{cls}}^u + \lambda \mathcal{L}_{\text{cls}}^s - \epsilon H(\mathbf{p})$. Combining the losses of the representation learning and the classifier learning, the overall loss of the baseline SimGCD is formulated by: $\mathcal{L}_{\text{baseline}} = \mathcal{L}_{\text{rep}} + \mathcal{L}_{\text{cls}}$.

3.1.3 One-vs-All Decomposition. Following DebGCD [14], we employ One-vs-All (OVA) classifiers for soft ID/OOD decomposition. For each known class $c \in \mathcal{Y}_l$, we pre-train a binary classifier $h_c : \mathbb{R}^d \to [0, 1]$ to distinguish class c from all others. For each unlabeled sample, we aggregate the per-class confidences into a single ID score $s_{ID}(\mathbf{x}^u) = \max_{c \in \mathcal{Y}_l} h_c(g(f(\mathbf{x}^u)))$, and derive soft weights:

$$w_{\text{old}}(\mathbf{x}^u) = s_{ID}(g(f(\mathbf{x}^u))), \quad w_{\text{new}}(\mathbf{x}^u) = 1 - s_{ID}(g(f(\mathbf{x}^u))), \tag{5}$$

Table 1: Comparison with state-of-the-art methods on generic and fine-grained benchmarks. All values are percentages.

	CUB			Stanford Cars			FGVC-Aircraft			CIFAR10			CIFAR100			ImageNet-100		
Methods	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New
k-means [19]	34.3	38.9	32.1	12.8	10.6	13.8	16.0	14.4	16.8	83.6	85.7	82.5	52.0	52.2	50.8	72.7	75.5	71.3
RS+ [7]	33.3	51.6	24.2	28.3	61.8	12.1	26.9	36.4	22.2	46.8	19.2	60.5	58.2	77.6	19.3	37.1	61.6	24.8
UNO+ [6]	35.1	49.0	28.1	35.5	70.5	18.6	40.3	56.4	32.2	68.6	**98.3**	53.8	69.5	80.6	47.2	70.3	**95.0**	57.9
GCD [28]	51.3	56.6	48.7	39.0	57.6	29.9	45.0	41.1	46.9	91.5	97.9	88.2	73.0	76.2	66.5	74.1	89.8	66.3
PromptCAL [34]	62.9	64.4	62.1	50.2	70.1	40.6	52.2	52.2	52.3	**97.9**	96.6	98.5	81.2	84.2	75.3	83.1	92.7	78.3
DCCL [23]	63.5	60.8	64.9	43.1	55.7	36.2	-	-	-	96.3	96.5	96.9	75.3	76.8	70.2	80.5	90.5	76.2
SimGCD [32]	60.3	65.6	57.7	53.8	71.9	45.0	54.2	59.1	51.8	97.1	95.1	98.1	80.1	81.2	77.8	83.0	93.1	77.9
SPTNet [30]	65.8	68.8	**65.1**	59.0	79.2	49.3	59.3	61.8	58.1	97.3	95.0	98.6	81.3	84.3	75.6	85.4	93.2	81.4
ProtoGCD [17]	63.2	68.5	60.5	53.8	73.7	44.2	56.8	62.5	53.9	97.3	95.3	98.2	81.9	82.9	80.0	84.0	92.2	79.9
DebGCD [14]	66.3	71.8	63.5	65.3	**81.6**	57.4	61.7	63.9	60.6	97.2	94.8	98.4	83.0	84.6	79.9	85.9	94.3	81.6
RPC (Ours)	**67.1**	**73.2**	64.8	**65.5**	81.0	**57.8**	**62.1**	**65.5**	**61.0**	97.6	95.2	**98.8**	**83.9**	**85.1**	**81.0**	**86.2**	93.8	**82.1**

where w_{old} indicates the likelihood of belonging to known classes and w_{new} to novel classes. These weights guide our bidirectional knowledge transfer: samples with high w_{old} participate in semantic pattern transfer from labeled data, while samples with high w_{new} contribute to relational consistency learning for novel discovery.

3.2 Relational Pattern Consistency (RPC)

Unlike prior GCD methods that treat labeled and unlabeled data as two disjoint domains, RPC explicitly couples them through bidirectional transfer (Fig. 1). Our approach rests on two insights: (i) labeled data contains rich semantic structure that can guide known-class learning beyond unreliable pseudo-labels, and (ii) same-category samples exhibit consistent behavioral signals when interacting with known-class prototypes.

3.2.1 Known-Class Preservation via Semantic Transfer. For known-class preservation, we transfer robust semantic patterns from labeled to unlabeled data through behavioral consistency under augmentation. We construct training batches by selecting ID-likely unlabeled samples based on OVA scores. For each batch of $\mathcal{B}_l$ labeled samples, we first sample μ unlabeled samples and compute $\mu_{ID} = \lfloor \mu \cdot \rho_{ID} \rfloor$, where ρ_{ID} is the estimated proportion of ID-class samples in the unlabeled set. We then select the top-μ_{ID} samples with highest w_{old} scores from these μ samples, resulting in the pattern: $[\mathbf{x}_1^l, \mathbf{x}_{1,1}^u, \ldots, \mathbf{x}_{1,\mu_{ID}}^u, \mathbf{x}_2^l, \mathbf{x}_{2,1}^u, \ldots, \mathbf{x}_{2,\mu_{ID}}^u, \ldots]$, where $\mathbf{x}_{i,j}^u$ denotes the j-th ID-class unlabeled sample paired with the i-th labeled sample, all sharing identical augmentations.

To enable semantic transfer from labeled to unlabeled samples, we apply confidence-weighted embedding [10]. Given the stacked batch embeddings $\mathbf{Z} \in \mathbb{R}^{Q \times d}$ where $Q = |\mathcal{B}_l|(1 + \mu_{ID})$, the fused embeddings $\mathbf{Z}' \in \mathbb{R}^{Q \times d}$ are obtained by:

$$\mathbf{Z}' = (\mathbf{I} + \mathbf{A})\mathbf{Z} \tag{6}$$

where $\mathbf{I} \in \mathbb{R}^{Q \times Q}$ is the identity matrix and $\mathbf{A}$ is the fusion matrix defined as follows. Following the circular shift design:

$$A_{ij} = \begin{cases} \alpha \cdot w_{\text{old}}(\mathbf{x}_i) & \text{if } j = (i-1) \bmod Q \\ -\alpha \cdot w_{\text{old}}(\mathbf{x}_i) & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

This design ensures that each unlabeled sample receives information from its preceding sample in the batch—either the paired labeled sample (when it is the first unlabeled sample in the group) or another unlabeled sample. The weighting by w_{old} so that only high-confidence ID samples are significantly affected.

With these fused embeddings $\mathbf{Z}'$, we enforce behavioral consistency between each labeled sample and its paired unlabeled samples:

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{B}_l|} \sum_{i=1}^{|\mathcal{B}_l|} \left\| \Delta(\mathbf{x}_i^l) - \frac{\sum_{j=1}^{\mu_{ID}} w_{\text{old}}(\mathbf{x}_{ij}^u) \cdot \Delta'(\mathbf{x}_{ij}^u)}{\sum_{j=1}^{\mu_{ID}} w_{\text{old}}(\mathbf{x}_{ij}^u)} \right\|_2^2, \tag{8}$$

where $\Delta(\mathbf{x}) = g(f(\mathcal{T}_w(\mathbf{x}))) - g(f(\mathcal{T}_s(\mathbf{x})))$ captures augmentation-induced behavioral changes, with $\mathcal{T}_w$ and $\mathcal{T}_s$ denoting weak and strong augmentations respectively. For labeled samples, $\Delta(\mathbf{x}^l)$ is computed on original features, while for unlabeled samples, $\Delta'(\mathbf{x}^u)$ is computed on fused features from $\mathbf{Z}'$.

3.2.2 Category Discovery via Relational Matching. For category discovery, we leverage a key insight: while novel classes are semantically distinct from known classes, samples from the same novel category share consistent patterns of similarity and dissimilarity with the known-class reference system. Specifically, similar distances to each known-class prototype serve as an implicit signature of category membership. We encode each sample's relational signature as:

$$\mathbf{r}(\mathbf{x}) = \left[\frac{f(\mathbf{x}) \cdot \mathbf{p}_1}{\|f(\mathbf{x})\|_2 \|\mathbf{p}_1\|_2}, \ldots, \frac{f(\mathbf{x}) \cdot \mathbf{p}_{C_L}}{\|f(\mathbf{x})\|_2 \|\mathbf{p}_{C_L}\|_2} \right] \in \mathbb{R}^{C_L}, \tag{9}$$

where $\{\mathbf{p}_c\}_{c=1}^{C_L}$ are known-class prototypes. This transforms each sample into coordinates within the known-class reference frame. We enforce relational consistency among feature-similar samples:

$$\mathcal{L}_{\text{new}} = \sum_{i,j \in \mathcal{B}_u} w_{\text{new}}(\mathbf{x}_i^u) \cdot w_{\text{new}}(\mathbf{x}_j^u) \cdot s_{ij} \cdot \|\mathbf{r}(\mathbf{x}_i^u) - \mathbf{r}(\mathbf{x}_j^u)\|_2^2, \tag{10}$$

where $s_{ij} = \exp(\langle f(\mathbf{x}_i^u), f(\mathbf{x}_j^u) \rangle / (\|f(\mathbf{x}_i^u)\|_2 \|f(\mathbf{x}_j^u)\|_2 \cdot \tau_u))$ weights pairs by feature similarity. This transforms novel discovery from ill-posed clustering into well-defined pattern matching—samples from the same novel category share consistent relationships with all known classes.

3.3 Training Objective

The complete RPC objective integrates baseline learning with our bidirectional consistency mechanisms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{baseline}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{new}} \tag{11}$$

Table 2: Ablation study on CUB and ImageNet-100.

Methods	CUB			ImageNet-100		
	All	Old	New	All	Old	New
w/o Embedding Fusion	64.9	70.1	62.4	84.0	90.6	81.2
w/o $\mathcal{L}_{\text{align}}$	65.6	71.0	63.6	84.6	91.4	81.5
w/o $\mathcal{L}_{\text{discover}}$	65.2	73.0	61.5	84.3	92.6	79.0
RPC (Ours)	**67.1**	**73.2**	**64.8**	**86.2**	**93.8**	**82.1**

where λ_1 and λ_2 balance known-class preservation and novel class relational consistency respectively.

4 Experiments

Datasets. We evaluate on six benchmarks: three fine-grained datasets from the Semantic Shift Benchmark (SSB)—CUB [31], Stanford Cars [11], and FGVC-Aircraft [20]—and two general benchmarks, CIFAR10/100 [12] and ImageNet-100 [27]. Following the GCD setting [28], we split categories into known ($\mathcal{Y}_l$) and novel classes, where 50% of known-class images form the labeled data $\mathcal{D}_L$ and all other images constitute the unlabeled pool $\mathcal{D}_U$. The total category count K ranges from 10 (CIFAR10) to 200 (CUB).

Implementation details. Following SimGCD [32], we adopt a ViT-B/16 backbone pre-trained with DINO [1], consistent with [28]. Image features, $f(\cdot)$, are extracted from the 768-dimensional [CLS] token. We fine-tune only the final block with a batch size of 128 over 200 epochs, using a learning rate of 0.1 with cosine decay. The balance factor λ is 0.35, τ_s=0.1, τ_u=0.07. We set λ_1=0.5, λ_2=0.3, and α=0.3 based on hyperparameter analysis.

Compared methods. We compare against three NCD methods (RS+ [7], UNO+ [6], k-means [19]) and seven GCD methods (GCD [28], PromptCAL [34], DCCL [23], SimGCD [32], SPTNet [30], ProtoGCD [17], DebGCD [14]). Our method builds on SimGCD's framework and DebGCD's OVA decomposition, adding bidirectional knowledge transfer.

4.1 Results and Discussion

Results analysis. Tab. 1 shows RPC achieves state-of-the-art performance across most benchmarks. Key findings: (1) Old ACC improves significantly (73.2% vs. DebGCD's 71.8% on CUB), validating semantic transfer. (2) New ACC gains 1.3% on CUB and 0.5% on ImageNet-100 via relational consistency. (3) Fine-grained datasets yield larger improvements (0.8% on CUB, 0.4% on Aircraft) than generic ones (0.3% on ImageNet-100).

Old-New trade-off. On ImageNet-100 and Stanford Cars, RPC improves New ACC while Old ACC is slightly lower than DebGCD (93.8% vs. 94.3% and 81.0% vs. 81.6%), due to embedding fusion mildly perturbing known-class boundaries to accommodate novel classes. On the three SSB benchmarks, both Old and New ACC improve simultaneously.

Effect of class composition. RPC shows consistent gains across novel-class ratios from 20% (CIFAR100) to 50% (CUB, FGVC-Aircraft, etc.): +1.1% New ACC on CIFAR100 and +1.3% on CUB over DebGCD.

Ablation Study Tab. 2 validates each component's contribution. Removing embedding fusion or $\mathcal{L}_{\text{align}}$ reduces known-class accuracy by 3.1% and 3.2% on CUB, confirming their role in semantic transfer. Without $\mathcal{L}_{\text{new}}$, novel-class accuracy drops 3.3% on CUB. All components prove essential.

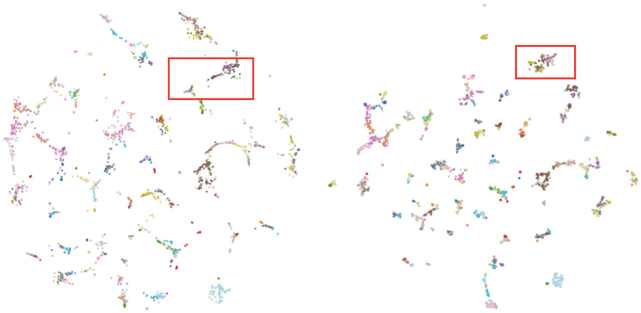

Figure 2: t-SNE visualization on CUB. (a) DebGCD and (b) RPC. Red boxes show RPC achieves tighter intra-class grouping and clearer inter-class boundaries.

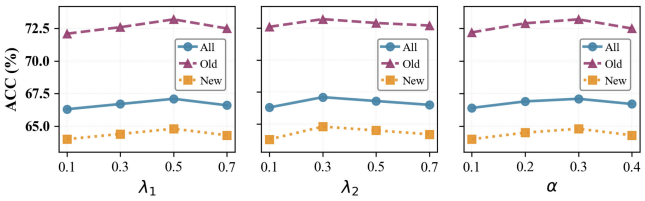

Figure 3: Impact of hyperparameters λ_1, λ_2, and α on CUB.

Table 3: Computational overhead comparison on CUB dataset.

Method	Params (M)	FLOPs (G)
SimGCD	92.25	17.60
RPC (Ours)	**93.54** (+1.40%)	**17.92** (+1.82%)

Hyperparameter Analysis. Fig. 3 shows RPC's robustness on CUB. Optimal values (λ_1=0.5, λ_2=0.3, α=0.3) balance preservation, discovery, and fusion. Performance degrades gracefully beyond these points, with overall variation within 1%.

Computational Efficiency & Feature Visualization. As shown in Tab. 3, RPC adds only 1.29M parameters (+1.40%) and 0.32G FLOPs (+1.82%) over SimGCD, introducing minimal overhead. Fig. 2 further shows t-SNE embeddings on CUB: RPC produces tighter intra-class grouping and clearer inter-class boundaries (red boxes), confirming better semantic structure preservation.

5 Conclusion

We proposed Relational Pattern Consistency (RPC) for GCD, enabling bidirectional knowledge transfer via embedding fusion and relational consistency. By treating labeled data as a semantic reference frame rather than merely a supervision source, RPC unifies known-class preservation with novel category discovery. Experiments on six benchmarks validate RPC's competitiveness, with particularly strong gains on fine-grained datasets where inter-class distinctions are subtle and relational cues prove critical. One limitation is the reliance on OVA classifiers for ID/OOD decomposition, whose accuracy directly affects downstream transfer quality. Extending RPC with automatic K estimation [36] and adaptive decomposition is a promising future direction.

References

[1] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. 2021. Emerging Properties in Self-Supervised Vision Transformers. In *ICCV*.

[2] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020. A simple framework for contrastive learning of visual representations. In *ICML*.

[3] Sua Choi, Dahyun Kang, and Minsu Cho. 2024. Contrastive mean-shift learning for generalized category discovery. In *CVPR*. 23094–23104.

[4] Jianan Fan, Dongnan Liu, Hang Chang, Heng Huang, Mei Chen, and Weidong Cai. 2024. Seeing unseen: Discover novel biomedical concepts via geometry-constrained probabilistic modeling. In *CVPR*. 11524–11534.

[5] Yixin Fei, Zhongkai Zhao, Siwei Yang, and Bingchen Zhao. 2022. XCon: Learning with Experts for Fine-grained Category Discovery. In *BMVC*.

[6] Enrico Fini, Enver Sangineto, Stéphane Lathuilière, Zhun Zhong, Moin Nabi, and Elisa Ricci. 2021. A Unified Objective for Novel Class Discovery. In *ICCV*.

[7] Kai Han, Sylvestre-Alvise Rebuffi, Sebastien Ehrhardt, Andrea Vedaldi, and Andrew Zisserman. 2021. AutoNovel: Automatically Discovering and Learning Novel Visual Categories. *PAMI* (2021). doi:10.1109/TPAMI.2021.3091944

[8] Zhenqi He, Yuanpei Liu, and Kai Han. 2025. Category discovery: An open-world perspective. *arXiv preprint arXiv:2509.22542* (2025).

[9] Ziming Huang, Xurui Li, Haotian Liu, Feng Xue, Yuzhe Wang, and Yu Zhou. 2025. Anomalyncd: Towards novel anomaly class discovery in industrial scenarios. In *CVPR*. 4755–4765.

[10] Zhe Huang, Xiaowei Yu, Dajiang Zhu, and Michael C Hughes. 2024. Interlude: Interactions between labeled and unlabeled data to enhance semi-supervised learning. *arXiv preprint arXiv:2403.10658* (2024).

[11] Jonathan Krause, Michael Stark, Jia Deng, and Li Fei-Fei. 2013. 3D object representations for fine-grained categorization. In *ICCV Workshops*.

[12] Alex Krizhevsky, Geoffrey Hinton, et al. 2009. *Learning multiple layers of features from tiny images*. Technical Report.

[13] Yu Liu, Yaqi Cai, Qi Jia, Binglin Qiu, Weimin Wang, and Nan Pu. 2024. Novel class discovery for ultra-fine-grained visual categorization. In *CVPR*. 17679–17688.

[14] Yuanpei Liu and Kai Han. 2025. Debgcd: Debiased learning with distribution guidance for generalized category discovery. *ICLR* (2025).

[15] Tingzhang Luo, Mingxuan Du, Jiatao Shi, Xinxiang Chen, Bingchen Zhao, and Shaoguang Huang. 2024. Contextuality helps representation learning for generalized category discovery. In *ICIP*. IEEE, 687–693.

[16] Tingzhang Luo, Yichao Liu, Yuanyuan Liu, Andi Zhang, Xin Wang, Yibing Zhan, Chang Tang, Leyuan Liu, and Zhe Chen. 2024. DIG-FACE: De-biased Learning for Generalized Facial Expression Category Discovery. *arXiv preprint arXiv:2409.20098* (2024).

[17] Shijie Ma, Fei Zhu, Xu-Yao Zhang, and Cheng-Lin Liu. 2025. Protogcd: Unified and unbiased prototype learning for generalized category discovery. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).

[18] Shijie Ma, Fei Zhu, Zhun Zhong, Xu-Yao Zhang, and Cheng-Lin Liu. 2024. Active generalized category discovery. In *CVPR*. 16890–16900.

[19] James MacQueen et al. 1967. Some methods for classification and analysis of multivariate observations. In *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability*.

[20] Subhransu Maji, Esa Rahtu, Juho Kannala, Matthew Blaschko, and Andrea Vedaldi. 2013. Fine-grained visual classification of aircraft. *arXiv preprint arXiv:1306.5151* (2013).

[21] Jianyuan Ni, Hao Tang, Syed Tousiful Haque, Yan Yan, and Anne HH Ngu. 2024. A survey on multimodal wearable sensor-based human action recognition. *arXiv preprint arXiv:2404.15349* (2024).

[22] Yunseok Oh and Dong-Wan Choi. 2025. FaceGCD: Generalized Face Discovery via Dynamic Prefix Generation. *arXiv preprint arXiv:2507.22353* (2025).

[23] Nan Pu, Zhun Zhong, and Nicu Sebe. 2023. Dynamic Conceptional Contrastive Learning for Generalized Category Discovery. In *CVPR*.

[24] Sarah Rastegar, Hazel Doughty, and Cees GM Snoek. 2023. Learn to Categorize or Categorize to Learn? Self-Coding for Generalized Category Discovery. *arXiv preprint arXiv:2310.19776* (2023).

[25] Yunhan Ren, Feng Luo, and Siyu Huang. 2025. Few-Shot Generalized Category Discovery With Retrieval-Guided Decision Boundary Enhancement. In *Proceedings of the 2025 International Conference on Multimedia Retrieval*. 1135–1144.

[26] Kihyuk Sohn, David Berthelot, Nicholas Carlini, Zizhao Zhang, Han Zhang, Colin A Raffel, Ekin Dogus Cubuk, Alexey Kurakin, and Chun-Liang Li. 2020. Fixmatch: Simplifying semi-supervised learning with consistency and confidence. In *NeurIPS*.

[27] Yonglong Tian, Dilip Krishnan, and Phillip Isola. 2020. Contrastive multiview coding. In *ECCV*.

[28] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. 2022. Generalized Category Discovery. In *CVPR*.

[29] Enguang Wang, Zhimao Peng, Zhengyuan Xie, Fei Yang, Xialei Liu, and Ming-Ming Cheng. 2025. Get: Unlocking the multi-modal potential of clip for generalized category discovery. In *CVPR*. 20296–20306.

[30] Hongjun Wang, Sagar Vaze, and Kai Han. 2024. Sptnet: An efficient alternative framework for generalized category discovery with spatial prompt tuning. In *ICLR*.

[31] Peter Welinder, Steve Branson, Takeshi Mita, Catherine Wah, Florian Schroff, Serge Belongie, and Pietro Perona. 2010. *Caltech-UCSD birds 200*. Technical Report. Computation & Neural Systems Technical Report.

[32] Xin Wen, Bingchen Zhao, and Xiaojuan Qi. 2023. Parametric Classification for Generalized Category Discovery: A Baseline Study. In *ICCV*.

[33] Yanan Wu, Zhixiang Chi, Yang Wang, and Songhe Feng. 2023. Metagcd: Learning to continually learn generalized category discovery. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 1655–1665.

[34] Sheng Zhang, Salman Khan, Zhiqiang Shen, Muzammal Naseer, Guangyi Chen, and Fahad Shahbaz Khan. 2023. Promptcal: Contrastive affinity learning via auxiliary prompts for generalized novel category discovery. In *CVPR*.

[35] Bingchen Zhao and Kai Han. 2021. Novel Visual Category Discovery with Dual Ranking Statistics and Mutual Knowledge Distillation. In *NeurIPS*.

[36] Bingchen Zhao, Xin Wen, and Kai Han. 2023. Learning Semi-supervised Gaussian Mixture Models for Generalized Category Discovery. In *ICCV*.

[37] Haiyang Zheng, Nan Pu, Wenjing Li, Nicu Sebe, and Zhun Zhong. 2024. Prototypical hash encoding for on-the-fly fine-grained category discovery. *NeurIPS* 37 (2024), 101428–101455.

[38] Haiyang Zheng, Nan Pu, Wenjing Li, Nicu Sebe, and Zhun Zhong. 2024. Textual knowledge matters: Cross-modality co-teaching for generalized visual class discovery. In *ECCV*. Springer, 41–58.

[39] Jiaying Zhou, Yang Liu, and Qingchao Chen. 2024. Novel class discovery in chest x-rays via paired images and text. In *AAAI*, Vol. 38. 7650–7658.

[40] Yuanhao Zuo, Yichao Liu, Xiwei Liu, and Tingzhang Luo. 2025. Linking Known and Unknown: Generalized Cross-Instance Feature Helps Category Discovery. In *ICASSP*. IEEE, 1–5.